Calc I Lecture 10: chain rule

Question: if we know f' and g', what is $\frac{d}{dx}(\underbrace{f \circ g}_{\text{composition of } f \text{ and } g.})$?

Example: $f(x) = e^x$, $g(x) = \sin(x)$

$$(f \circ g)(x) = f(g(x)) = e^{\sin(x)}$$

$$\frac{d}{dx}\left(e^{\sin(x)}\right) = ?$$

chain rule: if f and g are differentiable functions

$$\frac{d}{dx}(f \circ g)(x) = f'(g(x)) \cdot g'(x)$$

"derivative of outside function applied to inside function times derivative of inside function"

Back to example above:

$f(x) = e^x \Rightarrow f'(x) = \underline{e^x}$

$\underline{g(x) = \sin x} \Rightarrow g'(x) = \cos x$

$$\frac{d}{dx}\left(e^{\sin x}\right) = \frac{d}{dx}(f \circ g)(x) = \overbrace{f'(\sin x)}^{e^{\sin x}} \cdot \cos x = \boxed{e^{\sin x} \cdot \cos x}$$

Another perspective: Substitution

$$\frac{d}{dx}\left(e^{\overset{u}{\sin x}}\right)$$

Let $u = \sin x \Rightarrow \frac{du}{dx} = \cos x$

$$\frac{d}{dx}\left(e^{u}\right) = \underbrace{\frac{d}{du}\left(e^{u}\right)}_{e^{u}} \cdot \underbrace{\frac{du}{dx}}_{\cos x} = e^{u} \cdot \cos x$$

$$\underset{u=\sin x}{=} \boxed{e^{\sin x} \cdot \cos x}$$

$$\boxed{\frac{d}{dx}\left(f(u)\right) = \frac{df}{du} \cdot \frac{du}{dx}}$$ ← substitution perspective

Examples: Evaluate the following:

a) $\frac{d}{dx}\left(\sin\left(3x+2\right)\right)$ $\qquad \cos(3x+2) \cdot 3$

b) $\frac{d}{dx}\left(\cos^2(x)\right)$

c) $\frac{d}{dx}\left(e^{\sqrt{x}}\right)$

a) $\sin(3x+2) = f(g(x))$

$f(x) = \sin x \Rightarrow f' = \cos x$

$g(x) = 3x+2 \Rightarrow g' = 3$

$$\frac{d}{dx}\left(f \circ g\right) = f'(g(x)) \cdot g'(x)$$

$$= \cos(3x+2)\cdot 3 = \boxed{3\cos(3x+2)}$$

b) $\cos^2(x) = f(g(x))$

$= (\cos(x))^2$

$f(x) = x^2 \Rightarrow f'(x) = 2x$

$g(x) = \cos(x) \Rightarrow g'(x) = -\sin x$

$$\frac{d}{dx}(f\circ g) = \underbrace{f'(\underbrace{g(x)}_{\cos(x)})}_{2\cos(x)} \cdot \underbrace{g'(x)}_{-\sin(x)} = \boxed{-2\cos(x)\sin(x)}$$

Alternative approach: substitution

$u = \cos(x) \Rightarrow \frac{du}{dx} = -\sin x$

$\cos^2(x) = u^2$

$$\frac{d}{dx}(u^2) = \underbrace{\frac{d}{du}(u^2)}_{2u} \cdot \underbrace{\frac{du}{dx}}_{-\sin x} = -2u\sin(x)$$

$$= \boxed{-2\cos(x)\sin(x)}$$

c) $\frac{d}{dx}\left(e^{\sqrt{x}}\right)$

$e^{\sqrt{x}} \cdot \frac{1}{2}x^{-\frac{1}{2}}$

back to exponential functions

- $\frac{d}{dx}(e^x) = e^x$

- $\frac{d}{dx}(e^{2x}) =$

$u = 2x \Rightarrow \frac{du}{dx} = 2$

$\frac{d}{dx}(e^u) = \frac{d}{du}(e^u) \cdot \frac{du}{dx} = e^u \cdot 2$

$= \boxed{e^{2x} \cdot 2}$

- $\frac{d}{dx}(e^{cx}) = ce^{cx}$

- $\frac{d}{dx}(2^x) =$

idea: rewrite 2^x in base e

raise to the x

$2 = e^{\ln 2} \Rightarrow \underbrace{2^x} = (e^{\ln 2})^x = \underbrace{e^{(\ln 2)x}}$

$\Rightarrow \frac{d}{dx}(2^x) = \frac{d}{dx}(e^{\ln 2 \, x}) = \ln 2 \cdot \underbrace{e^{\ln 2 \, x}}_{2^x}$

$= \boxed{\ln 2 \cdot 2^x}$

- $\boxed{\dfrac{d}{dx}\left(b^x\right) = \ln b \cdot b^x}$

Question: what is $\dfrac{d}{dx}\left(x^x\right)$?

A little harder... discuss later.

But it's <u>not</u> $\ln x \cdot x^x$

<u>Example</u>: find $\dfrac{d}{dx}\left(\sec(\underbrace{3^x}_{u})\right)$

remember:

$\dfrac{d}{dx}(\sec x)$

$= \sec x \tan x$

$$\frac{d}{dx}(\sec(u)) = \frac{d}{du}(\sec(u)) \cdot \frac{du}{dx}$$

$$= \sec u \tan u \cdot u'$$

$$= \sec(3^x)\tan(3^x) \cdot \underbrace{\frac{d}{dx}(3^x)}_{\ln(3)\cdot 3^x}$$

Final answer: $\boxed{\sec(3^x)\tan(3^x) \cdot \ln(3) \cdot 3^x}$

Question: what if we compose 3 functions?

$$\frac{d}{dx}\left(\underbrace{f}_{\text{outside}}\circ \underbrace{g\circ h}_{\text{inside}}\right) = ?$$

chain rule $= f'(g\circ h)\cdot \underbrace{(\overset{\text{outside}}{g}\circ \overset{\text{inside}}{h})'}$

chain rule $= \boxed{f'(g\circ h)\cdot g'(h)\cdot h'}$

"chain"

Example: $\frac{d}{dx}\left(\sin\left(\sqrt{x^2-3}\right)\right) = ?$

outside: $\sin(x) \Rightarrow \frac{d}{dx}\sin(x) = \underline{\cos(x)}$

inside: $\underline{\sqrt{x^2-3}}$

$$\cos\left(\sqrt{x^2-3}\right)\cdot \underbrace{\frac{d}{dx}\left(\sqrt{x^2-3}\right)}$$

$\frac{d}{dx}(x^{\frac{1}{2}}) = \frac{1}{2}x^{-\frac{1}{2}}$

outside: $\sqrt{x} \Rightarrow \frac{d}{dx}\sqrt{x} = \boxed{\frac{1}{2\sqrt{x}}}$

inside: x^2-3

$$\frac{1}{2\sqrt{x^2-3}}\cdot \underbrace{\frac{d}{dx}(x^2-3)}_{2x}$$

Final answer: $\boxed{\cos\left(\sqrt{x^2-3}\right)\cdot \frac{1}{\cancel{2}\sqrt{x^2-3}}\cdot \cancel{2}x}$

More examples:

a) find $\dfrac{d}{dx}\left(\sqrt{1+\sqrt{1+\sqrt{x}}}\right)$

b) Suppose f is a function satisfying

$f(1)=1$, $f(2)=3$, $f'(1)=2$, $f'(2)=-1$

use this to find

$$\frac{d}{dx}\Big(f(1+f(1+2x))\Big) \quad \text{at} \quad x=0$$

Solutions

a) find $\dfrac{d}{dx}\left(\sqrt{1+\sqrt{1+\sqrt{x}}}\right)$

$$\frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}$$

$$\frac{d}{dx}\left(\sqrt{f(x)}\right) = \frac{1}{2\sqrt{\underbrace{f(x)}}} \cdot f'(x)$$

f

$$\frac{d}{dx}\left(\underbrace{\sqrt{1+\sqrt{1+\sqrt{x}}}}_{f}\right) = \frac{1}{2\underbrace{\sqrt{1+\sqrt{1+\sqrt{x}}}}_{f}} \cdot \underbrace{\frac{d}{dx}\left(1+\sqrt{1+\sqrt{x}}\right)}_{\frac{d}{dx}(1) + \frac{d}{dx}\sqrt{1+\sqrt{x}}}$$

$$\frac{d}{dx}(1) = 0$$

$$\frac{d}{dx}\underbrace{\sqrt{1+\sqrt{x}}}_{\text{inside}} = \frac{1}{2\sqrt{1+\sqrt{x}}} \cdot \underbrace{\frac{d}{dx}(1+\sqrt{x})}_{\frac{d}{dx}(1) + \frac{d}{dx}\sqrt{x}}$$

$$\frac{d}{dx}(1) = 0, \quad \frac{d}{dx}\sqrt{x} = \frac{1}{2\sqrt{x}}$$

Final answer: $\frac{1}{2\sqrt{1+\sqrt{1+\sqrt{x}}}} \cdot \left(0 + \frac{1}{2\sqrt{1+\sqrt{x}}} \cdot \left(0 + \frac{1}{2\sqrt{x}}\right)\right)$

$$= \boxed{\frac{1}{2\sqrt{1+\sqrt{1+\sqrt{x}}}} \; \frac{1}{2\sqrt{1+\sqrt{x}}} \; \frac{1}{2\sqrt{x}}}$$

b) Suppose f is a function satisfying

$f(1) = 1$, $f(2) = 3$, $f'(1) = 2$, $f'(2) = -1$

use this to find

$$\frac{d}{dx}\left(f(1+f(1+2x))\right) \quad \text{at } \underline{x=0}$$

$$\frac{d}{dx}\left(\underbrace{f}_{\text{outside}}(\underbrace{1+f(1+2x)}_{\text{inside}})\right)$$

$$= f'(1+f(1+2x)) \cdot \underbrace{\frac{d}{dx}(1+f(1+2x))}$$

$$\underbrace{\frac{d}{dx}(1)}_{=0} + \underbrace{\frac{d}{dx}(\underbrace{f}_{\text{outside}}(\underbrace{1+2x}_{\text{inside}}))}_{= f'(1+2x)\cdot \underbrace{\frac{d}{dx}(1+2x)}_{=2}}$$

$$= f'(1+f(1+2x)) \cdot (0 + f'(1+2x)\cdot 2)$$

$$= \boxed{f'(1+f(1+2x)) \cdot f'(1+2x) \cdot 2}$$

At $x=0$:

$1+2x \rightsquigarrow 1$

$$= f'(1+\overbrace{f(1)}^{1}) \cdot \overbrace{f'(1)}^{2} \cdot 2$$

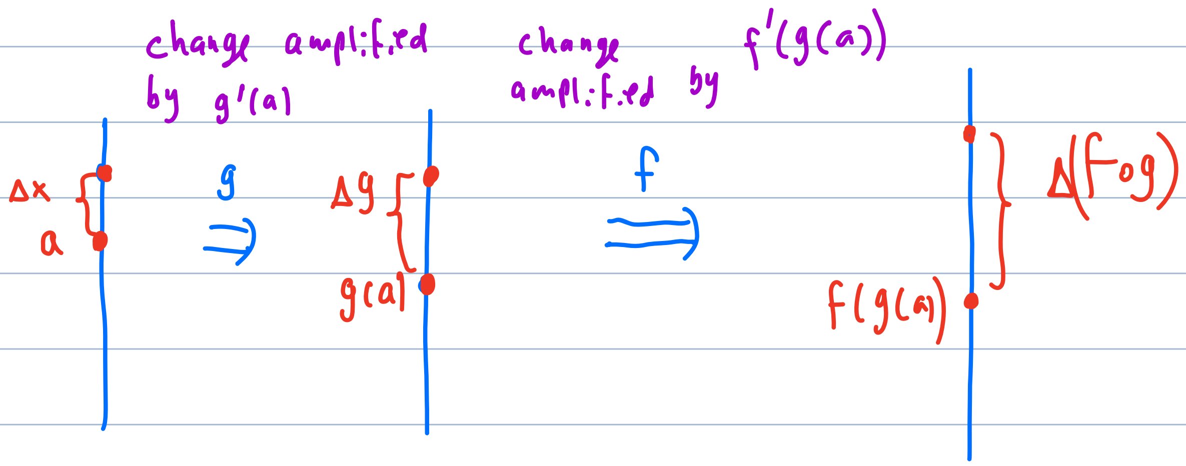
$$\boxed{f(1)=1, \quad f(2)=3, \quad f'(1)=2, \quad f'(2)=-1}$$ givens

$$= \underbrace{f'(1+1)}_{f'(2)=-1} \cdot 2 \cdot 2 = -1 \cdot 2 \cdot 2 = \boxed{-4}$$

intuition behind chain rule

Perspective: "amplifying change"

in general, $\Delta f \approx f'(a) \cdot \Delta x$ (change in x is "amplified" by factor of $f'(a)$)



$$\Delta x \overset{\cdot g'(a)}{\Longrightarrow} \underset{\Delta g \,\approx}{g'(a) \cdot \Delta x} \overset{\cdot f'(g(a))}{\Longrightarrow} \underset{\Delta(f\circ g)\,\approx}{f'(g(a)) \cdot g'(a) \cdot \Delta x}$$

$$\Delta(f \circ g) \approx f'(g(a)) \cdot g'(a) \cdot \Delta x \Rightarrow \frac{\Delta(f \circ g)}{\Delta x} \approx f'(g(a)) \cdot g'(a)$$

in conclusion: $\boxed{\frac{d}{dx}(f \circ g)(a) = f'(g(a)) \cdot g'(a)}$